

Data Acquisition for Automated Misinformation Detection

P.2: Data Sources and Preprocessing Plan

AI and Natural Language Processing

Rahul Mishra | MS Data Science, Northwestern University

April 13, 2026

Data Sources

Two publicly available datasets will support this project. The primary is the **LIAR benchmark** (Wang 2017), downloadable from https://www.cs.ucsb.edu/~william/data/liar_dataset.zip, which contains 12,836 labeled political statements from PolitiFact.com pre-split into training (10,269), validation (1,284), and test (1,283) sets. Each record carries twelve features: statement text, a six-class veracity label (*pants-fire* through *true*), speaker name, party affiliation, U.S. state, job title, a cumulative credit-history vector of per-class label counts, subject tags, and venue. The secondary dataset, **WELFake** (Verma, Agrawal, and Patel 2021), available at <https://www.kaggle.com/datasets/saurabhshahane/fake-news-classification>, provides 72,134 full-length news articles — 35,028 authentic and 37,106 fabricated — with article title, body text, and a binary label.

Preprocessing Plan

LIAR statements will be lowercased, Unicode-normalized, and tokenized with BERT WordPiece at 128 tokens; weighted cross-entropy loss will address the six-class label imbalance. The twelve structured metadata fields will be preserved separately for a feature-fusion phase. WELFake articles will have title and body concatenated with [SEP], truncated to 512 tokens, and split 80/10/10. Both datasets will be serialized as PyTorch *Dataset* objects for efficient, reproducible batched data loading.

Intended Use and Initial Methods

LIAR will serve as the primary multi-class benchmark; WELFake will enable binary classification and cross-dataset generalization tests. The core method is supervised fine-tuning of *bert-base-uncased* (Devlin et al. 2019) with a linear [CLS] classification head, benchmarked against a TF-IDF plus Logistic Regression baseline. A follow-up experiment fuses LIAR’s structured speaker metadata — party affiliation and credit-history vector — with the pooled transformer output via a feedforward fusion layer, testing whether contextual features overcome the text-only accuracy ceiling identified by Shu et al. (2017). Macro-averaged F1-score is the primary evaluation metric given class imbalance.

References

1. Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, Volume 1, pp. 4171–4186. Minneapolis, MN: Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>.
2. Shu, K., Sliva, A., Wang, S., Tang, J., and Liu, H. 2017. Fake News Detection on Social Media: A Data Mining Perspective. *ACM SIGKDD Explorations Newsletter* 19(1): 22–36. <https://doi.org/10.1145/3137597.3137600>.
3. Verma, P. K., Agrawal, P., and Patel, I. 2021. WELFake: Word Embedding over Linguistic Features for Fake News Detection. *IEEE Transactions on Computational Social Systems* 8(4): 881–893. <https://doi.org/10.1109/TCSS.2021.3068519>.
4. Wang, W. Y. 2017. “Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection. *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, Volume 2: Short Papers, pp. 422–426. Vancouver, Canada: Association for Computational Linguistics. <https://doi.org/10.18653/v1/P17-2067>.